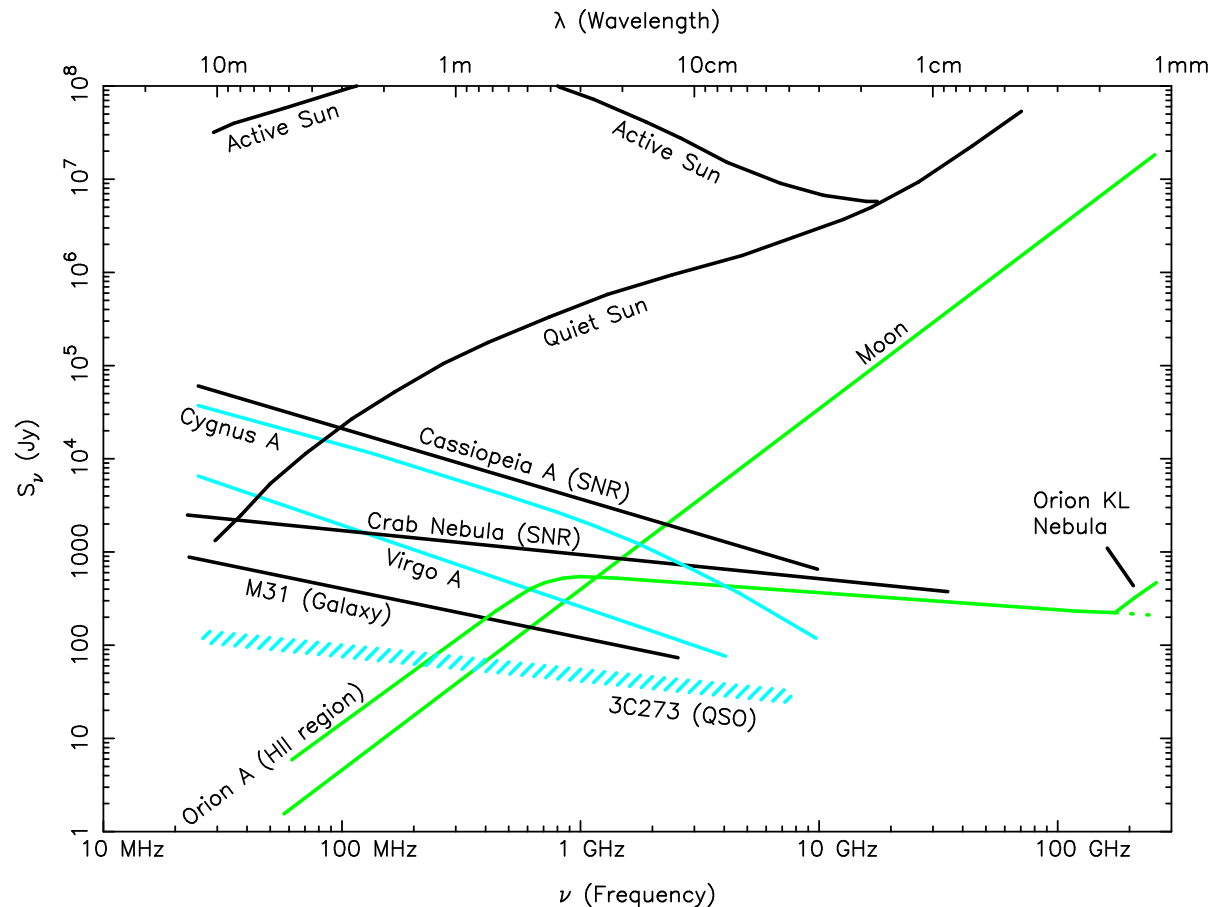

Lecture 8: Emission Mechanisms of Continuous Radiation

Chin-Fei Lee

`cflee@asiaa.sinica.edu.tw`

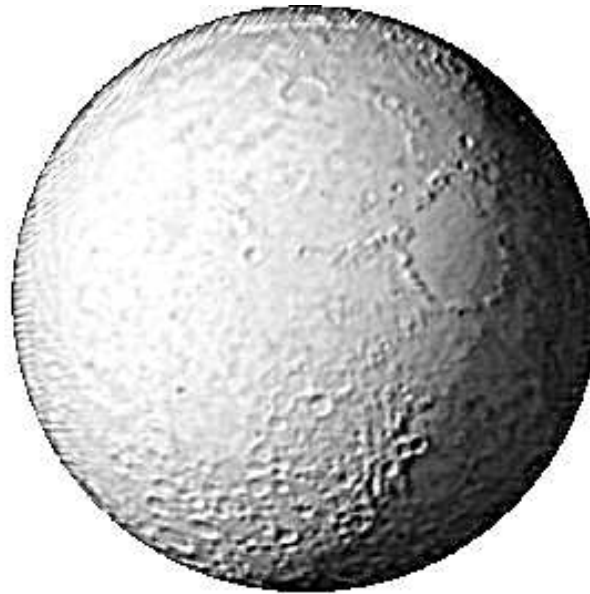
Nature of Radio Sources



The spectral distributions of various radio sources. The Moon, the quiet Sun and (at lower ν) the HII region Orion A (optically thin above 1 GHz) are examples of **Black Bodies**. Close to 300 GHz there is **additional emission from dust** in the molecular cloud Orion KL Nebula. The active Sun, SNRs such as Cassiopeia A, the radio galaxies Cygnus A, Virgo A and the Quasi Stellar radio source (Quasar or QSO) 3C273 are **nonthermal emission**. The hatching indicates rapid time variability. Quasars and radio galaxies are likely the same kind of objects, namely both with AGNs at the centers.

Thermal emission: Blackbody

The first examples of black bodies found were [solar system objects such as the Moon, planets and other solid bodies](#). At radio wavelengths, there is very little reflected sunlight so the radio emission observed from these objects is dominated by thermal emission from their surfaces. Their surfaces are heated by the Sun and radiate in [infrared and radio](#). If the Sun were suddenly switched off, the planets would still remain radio sources for a long time, slowly fading as they cooled.



[Thermal emission from the Moon at \$\lambda = 850 \mu\text{m}\$](#) . The darker right edge ($\sim 100 \text{ K}$) of the Moon is not being illuminated by the Sun, but it still emits radio waves because it does not cool to absolute zero during the lunar night. The brighter side is hotter ($200\text{--}300 \text{ K}$) and emits more emission.

Thermal emission: Blackbody

For such objects, $\tau = \infty$, so

$$I_\nu = B_\nu(T_s) \equiv \frac{2h\nu^3}{c^2} \frac{1}{e^{h\nu/kT} - 1}$$

and the radio telescope measures the temperature at a depth corresponding to a λ or so. This temperature, referred to as **surface temperature T_s** , is nearly constant with frequency. In the Rayleigh-Jeans limit, $B_\nu^{RJ} \approx \frac{2\nu^2}{c^2} kT_s$ and thus

$$I_\nu = \frac{2\nu^2}{c^2} kT_s \quad (\propto \nu^2 \text{ as seen in the spectrum})$$

and the brightness temperature is

$$T_b \equiv \frac{c^2}{2k\nu^2} I_\nu = T_s$$

The measured main beam temperature is then

$$T_{\text{MB}} = fT_b = fT_s \quad \text{where} \quad f \equiv \frac{\theta_s^2}{\theta_s^2 + \theta_b^2}$$

where f is the beam filling factor. Once θ_s and thus f are known, T_s can be derived accurately.

Thermal Radiation of Dust

The InterStellar Medium (ISM) is mainly composed of matter in two physical states: [gas and dust](#). Dust absorbs the energy of nearby stars in the ultra violet and optical domains and radiates mainly in the infrared and submillimetre domain. Although it represents only 1% of the gas mass, it is responsible for around 30% of the total energy radiated by the Galaxy. It is therefore the most intense astrophysical radiation source at frequencies above 100 GHz.

In the case of molecular clouds, the emission is from small dust particles (with the sizes $\sim 0.1 - 10 \mu\text{m}$). Assuming a constant dust temperature T_{dust} and a RJ limit, the radiative transfer in brightness temperature is

$$T_b = T_{b0}e^{-\tau_{\text{dust}}} + T_{\text{dust}}(1 - e^{-\tau_{\text{dust}}}), \quad h\nu \ll kT$$

where $T_{b0} = 2.7 \text{ K}$ from the microwave background radiation. Usually $T_{\text{dust}} \gg T_{b0}$ and thus

$$T_b \approx T_{\text{dust}}(1 - e^{-\tau_{\text{dust}}})$$

Thermal Radiation of Dust

For $\lambda > 100 \mu\text{m}$ (or 0.1 mm in radio regime), the dust opacity is (Mezger et al. 1990)

$$\tau_{\text{dust}} = 7 \times 10^{-21} \frac{Z}{Z_{\odot}} b N_{\text{H}} \lambda^{-2}$$

where λ is in μm , N_{H} is in cm^{-2} , Z is the metallicity as a ratio of that of the sun $Z_{\odot}$. $b \sim 1.9$ for moderate density gas and 3.4 for dense gas. In the clouds, $\tau_{\text{dust}} \ll 1$, and the brightness temperature is

$$T_b \approx T_{\text{dust}} \tau_{\text{dust}}$$

and the flux density is

$$S = \frac{2kT_b}{\lambda^2} \Delta\Omega = \frac{2kT_{\text{dust}} \tau_{\text{dust}}}{\lambda^2} \Delta\Omega \propto \lambda^{-4} \propto \nu^4$$

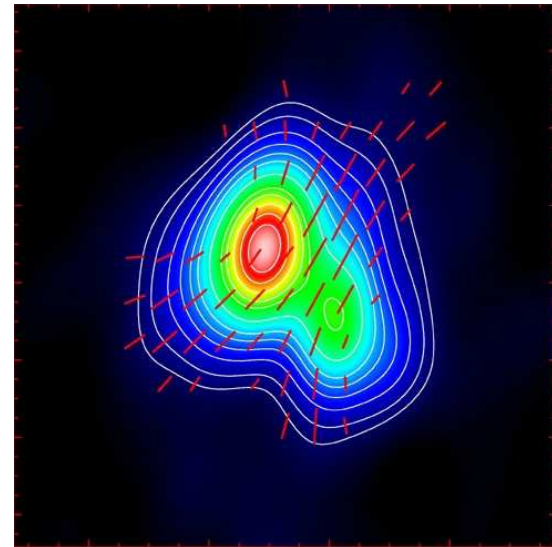
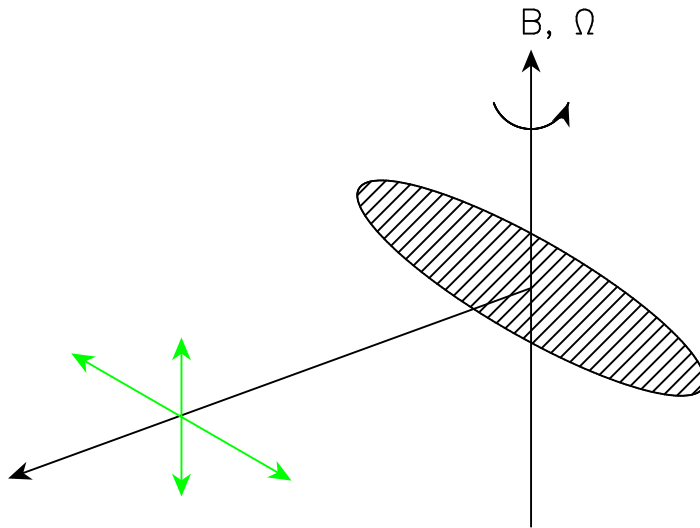
Measuring S , we can determine the column density of N_{H} . If S in mJy, λ in mm, the source in FWHP sizes, θ in arc seconds, then

$$N_{\text{H}} = 1.93 \times 10^{24} \frac{S}{\theta^2} \frac{\lambda^4}{Z/Z_{\odot} b T_{\text{dust}}} \quad (\text{cm}^{-2})$$

It appears that cold dust (10-30 K) makes up most of the mass of dust and by implication traces cold interstellar gas (assuming a gas to dust ratio of about 100) in our galaxy.

Dust Polarization

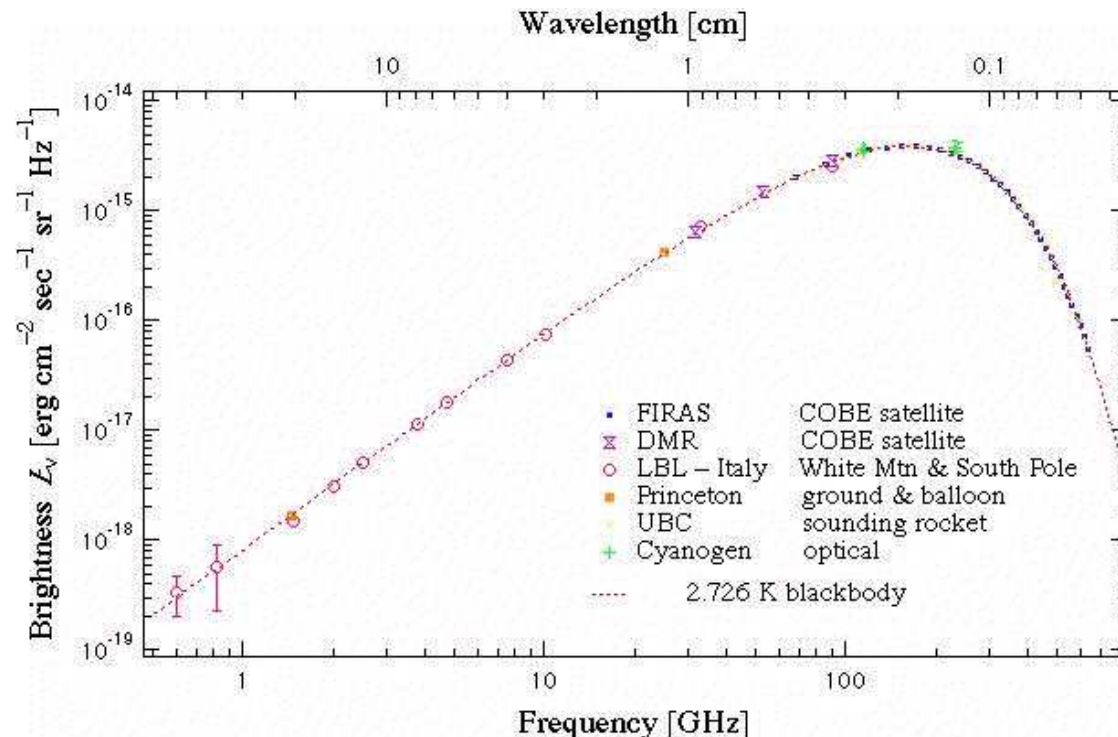
If the dust grains are **non-spherical**, and some process aligns the grains, one would expect the thermal emission to be polarized. Eg., if the grains have an electric dipole and as more dipoles are created along the largest axis of the grains, they are aligned by a B field (via torque) with the largest axis aligned perpendicular to the B field, then the thermal emission will be more intense perpendicular to the B field $\Rightarrow$ **measuring the direction of B field**.



SMA has done some really nice dust polarization observations toward star forming cores.

Cosmic microwave background

The most famous black body radiation is the 2.7 K microwave background radiation, discovered in 1965. According to the Big Bang scenario, the cosmic microwave background radiation is a relic of the earliest phase of the Universe. The difficulty in detecting it was not due to its weakness, but rather due to the fact that the radiation is present in all directions, so that scanning a telescope over the sky will not lead to a detection. The actual discovery was due to exact measurements of the noise temperature from the sky, the receiver and the ground, compared to the temperature of a helium cooled load. With more advanced instruments, the spectral energy distribution (SED) has been measured beautifully.



HII region: Ionized Gas Cloud



HST Image: Orion Nebula (a.k.a M42).

An H II region is a cloud of glowing gas and plasma, sometimes several hundred light-years across, in which star formation is taking place. Young, hot, blue stars which have formed from the gas emit copious amounts of ultraviolet light, ionizing the nebula surrounding them. H II regions are named for the large amount of ionized atomic hydrogen (H^+) they contain.

H II regions may give birth to thousands of stars over a period of several million years. In the end, supernova explosions and strong stellar winds from the most massive stars in the resulting star cluster will disperse the gases of the H II region, leaving behind a cluster such as the Pleiades.

The radio continuum in the HII region is from the electrons accelerated in the electron-ion interaction.

Radiation from Accelerated Electron

An accelerated Electron $\Leftrightarrow$ temporal variation in current as in Hertz dipole $\Rightarrow$ Radiation

From the Hertz dipole, we have (assuming $\mu \approx 1$ and $\epsilon \approx 1$) the radiation field

$$E_{\theta} = -i \frac{k I \Delta l}{c} \frac{\sin \theta}{r} \exp[-i(\omega t - kr)]$$

the other components are zero. Now assuming $v(t) = v_0 e^{-i\omega t}$ then $\dot{v} = -i\omega v$ and thus

$$I \Delta l = -ev = \frac{e\dot{v}}{i\omega} = \frac{e\dot{v}}{ikc}$$

Substituting this into the above eq., we have

$$E_{\theta} = -\frac{e\dot{v}}{c^2} \frac{\sin \theta}{r} \exp[-i(\omega t - kr)]$$

The electric field induced at the position of the observer by a charge at the distance r moving with the velocity $v(t)$ can be given by an expression that contains the acceleration $\dot{v}$ of the charge e at the retarded time.

Radiation from Accelerated Electron

The Poynting flux, that is, the power per surface area and steradian emitted into the direction (ϑ, φ) is

$$|S| = \frac{c}{4\pi} \mathbf{E}^2 = \frac{1}{4\pi} \frac{e^2 \dot{v}^2}{c^3} \frac{\sin^2 \theta}{r^2}$$

Integrating this over the full sphere results in the total amount of power radiated when a charge is accelerated (as for Hertz dipole):

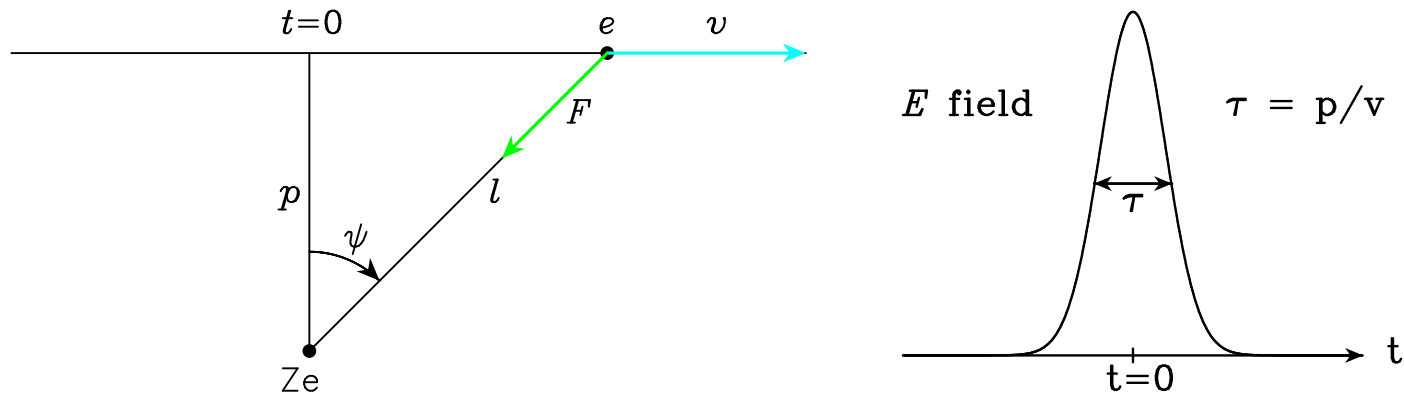
$$P(t) = \int |S| r^2 d\Omega = \frac{2}{3} \frac{e^2 \dot{v}^2(t)}{c^3}$$

The total amount of energy emitted during the whole encounter is

$$W = \int_{-\infty}^{\infty} P(t) dt = \frac{2}{3} \frac{e^2}{c^3} \int_{-\infty}^{\infty} \dot{v}^2 dt$$

Now we need to ask: [How is the electron accelerated and what is \$\dot{v}\$?](#)

Bremsstrahlung: electron + ion



Considering an electron moving past an ion of charge Ze with large distance so that the path (hyperbola in reality) can be approximated by a straight line. Let $l = p$ and $\psi = 0$ at $t = 0$ and thus

$$l = \frac{p}{\cos \psi}$$

The acceleration of the electron is given by the **Coulomb's Force**

$$F = m\dot{\mathbf{v}} = -\frac{Ze^2}{l^2} \frac{\mathbf{l}}{l} \quad \Rightarrow \quad \dot{\mathbf{v}} = -\frac{Ze^2}{ml^2} \frac{\mathbf{l}}{l}$$

For large p , F and thus $\dot{\mathbf{v}}$ and E field strength is large when ψ is small, in which the acceleration is mainly directly perpendicular to the orbit (path). In that case,

$$\dot{v} \simeq |\dot{\mathbf{v}}| \cos \psi = -\frac{Ze^2}{mp^2} \cos^3 \psi$$

Acceleration by an ion

Thus the total amount of energy emitted during the ion-electron interaction is

$$W = \frac{2}{3} \frac{e^2}{c^3} \int_{-\infty}^{\infty} \dot{v}^2 dt = \frac{4}{3} \frac{Z^2 e^6}{c^3 m^2 p^4} \int_0^{\infty} \cos^6 \psi \, d\psi$$

To evaluate the integral, we need to derive the dependence of ψ on t . [Kepler's law of areas](#): for any motion governed by a central force, the area swept per unit time by the line connecting the ion and electron is constant. Since the infinitesimal area is given by

$$dA = \frac{1}{2} l^2 d\psi$$

we have

$$\frac{dA}{dt} = \frac{1}{2} l^2 \frac{d\psi}{dt} = \text{constant} = \frac{1}{2} p v$$

with the definitions that $l = p$ and $l \frac{d\psi}{dt} = v$ at $t = 0$. Indeed $|v| \simeq \text{const}$ during the interaction. Thus,

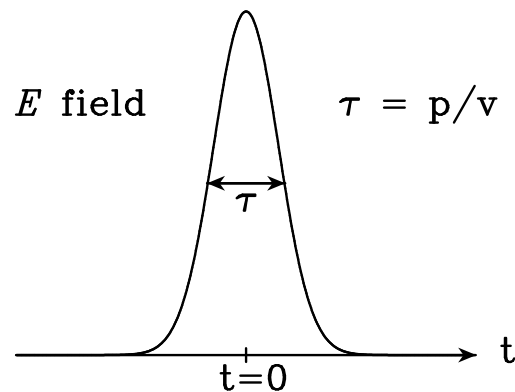
$$dt = \frac{l^2}{vp} d\psi = \frac{p}{v \cos^2 \psi} d\psi$$

Acceleration by an ion

Thus, the total amount of energy emitted is

$$W = \frac{4}{3} \frac{Z^2 e^6}{c^3 m^2 p^4} \frac{p}{v} \int_0^{\pi/2} \cos^4 \psi \, d\psi = \frac{\pi}{4} \frac{Z^2 e^6}{c^3 m^2 p^3} \frac{1}{v}$$

Now, what is the radiated frequency for this energy? The ion-electron interaction is transient and only results in a pulse that can be considered as a wave packet consisting of a group waves of frequency ω :



$$E(t) = \int_{-\infty}^{\infty} A(\omega) e^{-i\omega t} d\omega$$

and the wave amplitude $A(\omega)$ is given by

$$A(\omega) = \frac{1}{2\pi} \int_{-\infty}^{\infty} E(t) e^{i\omega t} dt$$

Here τ can be considered as the collision time. Again Fourier Transform!

Acceleration by an ion

The derivation of $A(\omega)$ is complicated. The result can be approximated as a step function:

$$A(\omega) = \begin{cases} 1 & \text{for } \omega \leq \omega_g \equiv \frac{3\pi^2}{16} \frac{v}{p} = 1.851 \frac{v}{p} \\ 0 & \text{otherwise} \end{cases}$$

The resulting frequency can reach up to $1.851 \frac{v}{p}$, where $\frac{v}{p} = \frac{1}{\tau}$ ($\Delta t \Delta E \gtrsim \hbar \Rightarrow \Delta \omega \gtrsim \frac{1}{\tau} \sim \frac{v}{p}$).

Thus for a given collisions parameters (p, v) , the spectrum can be approximated as follow

$$P_\nu(p, v) = \begin{cases} \frac{W}{\nu_g} = \frac{8}{3} \frac{Z^2 e^6}{c^3 m^2} \frac{1}{p^2 v^2} & \text{for } \nu \leq \nu_g \equiv \frac{\omega_g}{2\pi} = \frac{3\pi}{32} \frac{v}{p} \\ 0 & \text{otherwise} \end{cases}$$

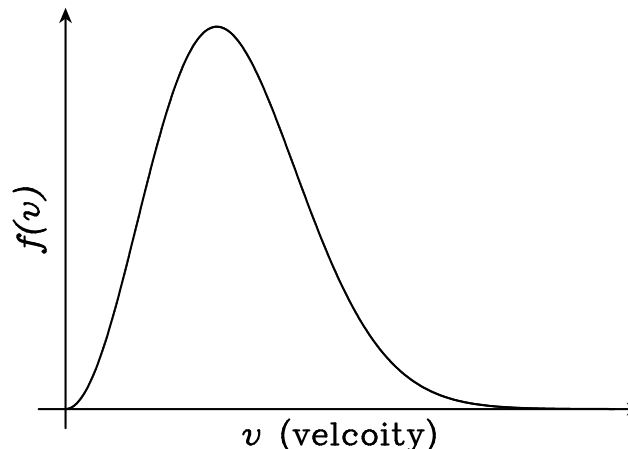
Radiation of Ionized Gas Cloud

Consider an ionized gas cloud, in which electrons have a distribution of p and v . In a thermalized medium, the **probability** dP that an electron has velocity in the velocity range $d^3\mathbf{v}$ is

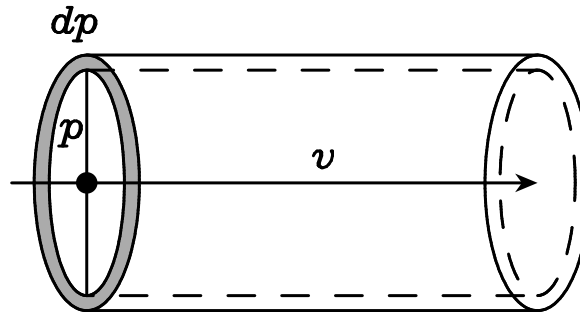
$$dP \propto e^{-E/kT} d^3\mathbf{v} = \exp\left(-\frac{mv^2}{2kT}\right) d^3\mathbf{v}$$

where m is the electron mass. Since $d^3\mathbf{v} = 4\pi v^2 dv$ for an isotropic distribution of velocities, the probability that an electron has a speed in the speed range dv is a **Maxwellian distribution**:

$$dP = \frac{4v^2}{\sqrt{\pi}} \left(\frac{m}{2kT}\right)^{3/2} \exp\left\{-\frac{mv^2}{2kT}\right\} dv \equiv f(v) dv$$



Radiation of Ionized Gas Cloud



The number of electrons in a unit volume of space with a given velocity $v = \sqrt{\mathbf{v}^2}$ that will pass by a given ion with a collision parameter between p and $p + dp$ is

$$2\pi p dp v N_e f(v) dv \quad \text{s}^{-1}$$

So number of collisions per second with collision parameters between p and $p + dp$ and velocities between v and $v + dv$ is

$$dN(p, v) = 2\pi N_i N_e v p f(v) dv dp \quad \text{cm}^{-3} \text{ s}^{-1}$$

where N_i is the number of ions per unit volume.

Radiation of Ionized Gas Cloud

As a result of these collisions, a power of $4\pi j_\nu d\nu$ will be radiated in the frequency interval ν to $\nu + d\nu$ given by

$$4\pi j_\nu d\nu = P_\nu(p, \nu) dN(p, \nu) d\nu$$

Substituting for P_ν and dN and integrating ν from 0 to ∞ and p from p_1 to p_2 , then we have

$$4\pi j_\nu = \int_{p_1}^{p_2} \int_0^\infty \frac{8}{3} \frac{Z^2 e^6}{c^3 m^2} \frac{1}{p^2 \nu^2} 2\pi N_i N_e \nu p f(\nu) d\nu dp$$

where p_2 should be the mean distance between the ions or the Debye length (over which significant charge separation can occur) in the plasma. p_1 is from the quantum mechanics uncertainty principle. Thus, the emission coefficient is

$$j_\nu = \frac{4}{3} \frac{Z^2 e^6}{c^3} N_i N_e \sqrt{\frac{2}{\pi m^3 k T}} \ln \frac{p_2}{p_1}$$

The absorption coefficient in Rayleigh-Jeans limits is

$$\alpha_\nu = \frac{j_\nu}{B_\nu(T)} = \frac{j_\nu}{\frac{2\nu^2}{c^2} k T} = \frac{4}{3} \frac{Z^2 e^6}{c \nu^2} N_i N_e \sqrt{\frac{1}{2\pi (m k T)^3}} \ln \frac{p_2}{p_1}$$

Radiation of Ionized Gas Cloud

The optical depth is

$$\tau_\nu = \int_0^s \alpha_\nu ds$$

which is integrated over the cloud and s is the size of the cloud. If the cloud is an HII region, $Z = 1$, $N_I \simeq N_e$. Assuming T is constant along the line of sight, then

$$\tau_\nu = 8.235 \times 10^{-2} \left(\frac{T}{\text{K}}\right)^{-1.35} \left(\frac{\nu}{\text{GHz}}\right)^{-2.1} \left(\frac{\text{EM}}{\text{pc} \cdot \text{cm}^{-6}}\right) a(\nu, T)$$

where $a(\nu, T) \simeq 1$ and the **emission measure EM** is defined as

$$\frac{\text{EM}}{\text{pc} \cdot \text{cm}^{-6}} = \int_0^{s/\text{pc}} \left(\frac{N_e}{\text{cm}^{-3}}\right)^2 d\left(\frac{s}{\text{pc}}\right)$$

HII regions

HII can be assumed to be in LTE with constant temperature. Thus, the radiative transfer is:

$$I_\nu(\tau_\nu) = I_\nu(0)e^{-\tau_\nu} + B_\nu(1 - e^{-\tau_\nu})$$

Here $I_\nu(0)$ is the microwave background radiation and is negligible compared to B_ν at $T \approx 8000$ K for typical HII region. Thus, we have

$$I_\nu(\tau_\nu) = B_\nu(1 - e^{-\tau_\nu})$$

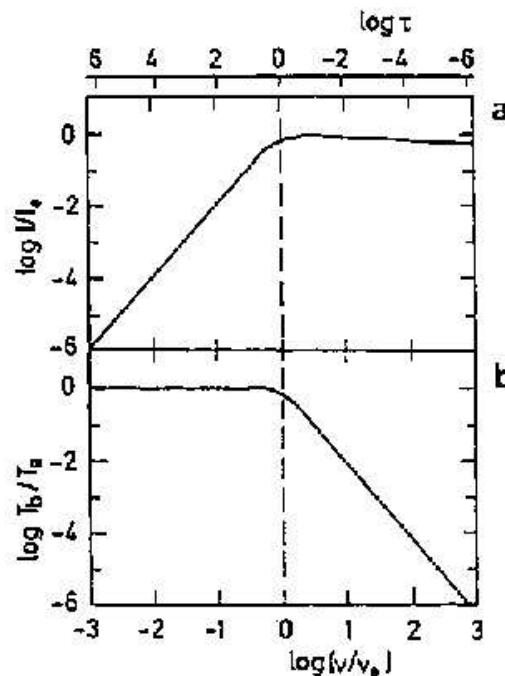
$$I_\nu(\tau_\nu) = \begin{cases} B_\nu = \frac{2\nu^2}{c^2}kT \propto \nu^2 & \text{for } \tau_\nu \gg 1 \\ B_\nu\tau_\nu \propto \nu^{-0.1} & \text{for } \tau_\nu \ll 1 \end{cases}$$

And $\tau_\nu = 1$ at the frequency ν_0 which is given by

$$\frac{\nu_0}{\text{GHz}} = 0.3045 \left(\frac{T}{\text{K}} \right)^{-0.643} \left(\frac{a(\nu, T)\text{EM}}{\text{pc} \cdot \text{cm}^{-6}} \right)^{0.476}$$

This frequency is called the turn over frequency at which the frequency dependency of I_ν changes from ν^2 to $\nu^{-0.1}$. For typical bright HII regions like M42, ν_0 is of the order of $\simeq 1$ GHz, EM $\approx 4 \times 10^6$ pc cm $^{-6}$ and $T \approx 8000$ K.

HII regions



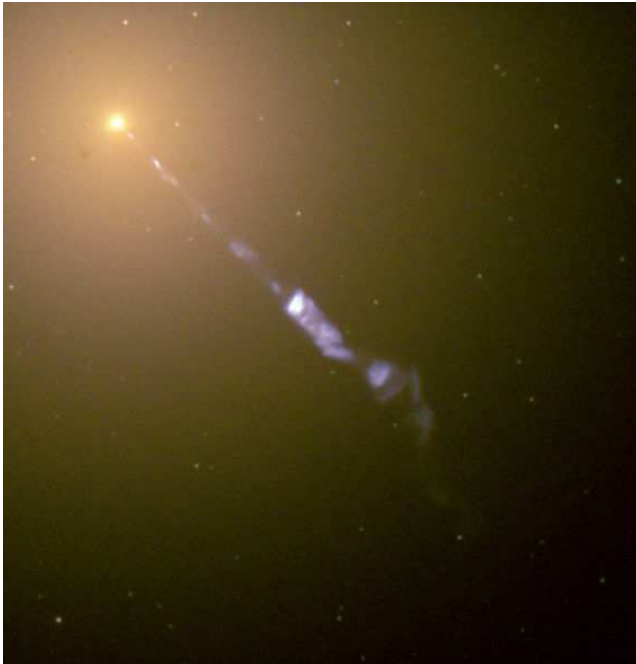
Note the brightness temperature $T_b = \frac{c^2}{2\nu^2 k} I_\nu$, Thus,

$$T_b(\tau_\nu) = \begin{cases} T & \text{for } \tau_\nu \gg 1 \\ T\tau_\nu \propto \nu^{-2.1} & \text{for } \tau_\nu \ll 1 \end{cases}$$

From the observations, we measure T and ν_0 and then we can derive EM. If the size of the HII region is known, we can also derive the electron number density N_e .

Nonthermal Radiation

In 1950, Alfvén and Herlofson proposed that the nonthermal radiation is emitted by [relativistic electrons moving in intricately "tangled" magnetic fields](#) of extended coronas believed to surround certain kinds of stars. The radiation was given the name “synchrotron radiation” since it was first discovered to be emitted by [electrons in a synchrotron accelerator](#). Synchrotron radiation is similar to [cyclotron radiation](#), but from ultrarelativistic electrons.



Hubble Space Telescope image of a 5000 light-year long jet being ejected from the active nucleus (AGN) of the active galaxy M87 (a radio galaxy, a.k.a Virgo A). The synchrotron radiation of the jet (blue) contrasts with the starlight from the host galaxy (yellow). The synchrotron radiation is emitted by high-energy electrons spiraling along magnetic field lines. It was first detected in 1956 by Geoffrey R. Burbidge in M87, confirming a prediction by Hannes Alfvén and Nicolai Herlofson in 1950, and Iosif S. Shklovskii in 1953.

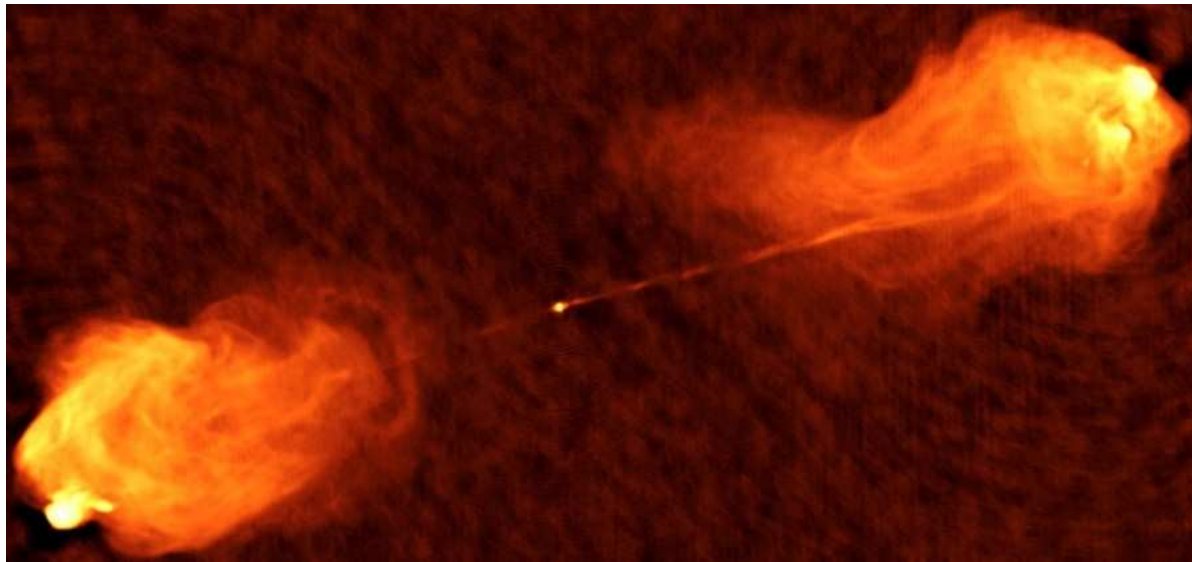
AGNs have two classes: radio-quiet and radio-loud.

In the radio-loud objects a contribution from the jet(s) and the lobes they inflate dominates the luminosity of the AGN, at least at radio wavelengths but possibly at some or all others.

Radio-quiet objects are simpler since jet and jet-related emission can be neglected.

Cygnus A

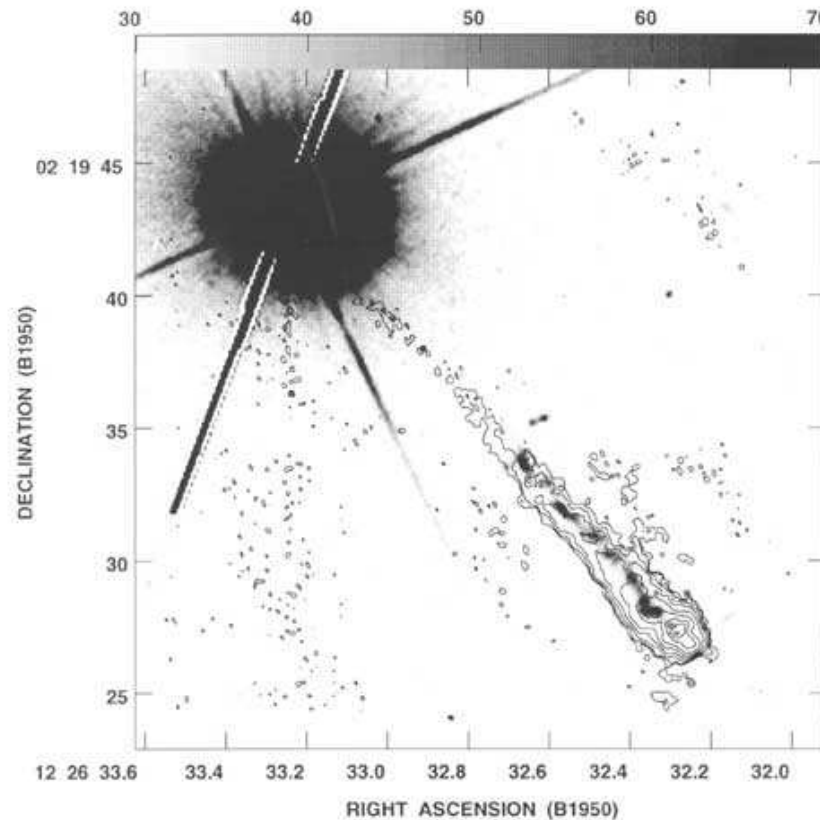
The strongest extragalactic radio source in the sky is the radio galaxy Cygnus A. The 1954 identification of this source with an extremely distant (redshift $z \approx 0.057$, corresponding to a distance $D \sim 240$ Mpc) galaxy stunned radio astronomers, who immediately recognized that such a luminous radio source ($\approx 10^{45} \text{ erg s}^{-1} = 10^{38} \text{ W}$) could be detected almost anywhere in the universe. The angular extent of this source, about 100 arcsec, implies a linear extent of about 100 kpc, much larger than the host galaxy. The energy source is clearly not stars. Gravitational energy released by matter accreting onto a supermassive ($M \sim 10^9 M_{\odot}$) black hole in the center of the host galaxy powers this and other luminous extragalactic radio sources.



A high-resolution VLA image of the radio source Cygnus A. The bright central component is thought to coincide with a supermassive black hole that accelerates the relativistic electrons along two jets that terminate in lobes well outside the host galaxy.

Nonthermal Radiation

The bright radio source 3C 273 was identified with the first quasar at an even higher redshift, $z=0.16$. Such quasars appear to be radio galaxies in an especially active state, when visible light from the region of the black hole overwhelms the starlight from the host galaxy and makes the quasar look like a bright star.



This HST gray-scale image of the quasar 3C273 shows radio contours superimposed on the optical jet emission.

Synchrotron Radiation

Since the electrons that emit the synchrotron radiation are highly relativistic, [Lorentz transformations](#) (special theory of relativity) must be used to describe their motion.

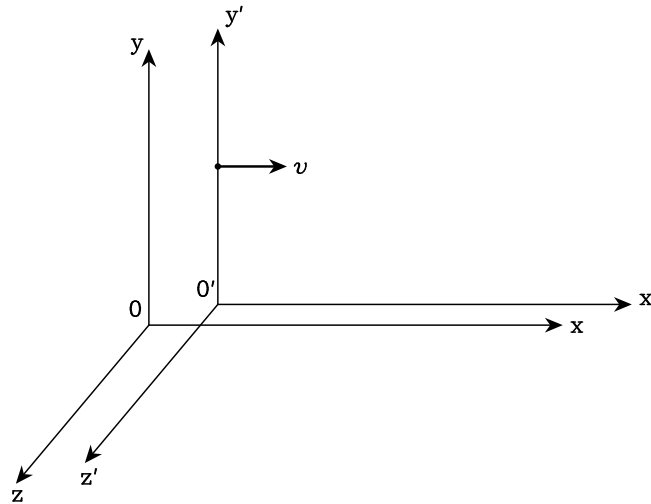
The Lorentz transformation converts between two different observers' measurements of space and time, where one observer is in constant motion with respect to the other. With this, we can derive our solution in one frame (i.e., an inertial frame moving with the electron and thus nonrelativistic physics can be adopted) and convert it to another frame (observer's frame).

There are two postulates in special theory of relativity:

1. The laws of nature are the same in any two frames of reference that are in uniform relative motion
2. The speed of light, c , is constant in all such frames.

Lorentz Transformation

Two inertial frames K and K' with a relative velocity v along the x-axis:



The origins are assumed to coincide at $t = 0$. If a pulse of light is emitted at the origin at $t = 0$, each observer will see an expanding sphere centered on his own origin because the speed of light is c and constant:

$$x^2 + y^2 + z^2 - c^2 t^2 = 0 \qquad x'^2 + y'^2 + z'^2 - c^2 t'^2 = 0$$

In another word, the evolutions of space and time are bonded by the speed of light $\Rightarrow$ [4-dimensional spacetime](#) and lengths and times must be transformed together from one frame to another. If space is homogeneous, then the Lorentz transformation must be a linear transformation.

Lorentz Transformation

The transformation can be thought of some kind of rotation in spacetime with

$$\begin{bmatrix} ct \\ x \\ y \\ z \end{bmatrix} = \begin{bmatrix} \gamma & \beta\gamma & 0 & 0 \\ \beta\gamma & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} ct' \\ x' \\ y' \\ z' \end{bmatrix}$$

Or

$$\begin{aligned} t &= \gamma(t' + \beta \frac{x'}{c}) \\ x &= \gamma(x' + vt') \\ y &= y' \\ z &= z' \end{aligned}$$

where

$$\beta = \frac{v}{c} \quad \text{and} \quad \gamma^2 - \beta^2 \gamma^2 = 1 \Rightarrow \gamma = (1 - \beta^2)^{-1/2}$$

As a result, the lengths and times are distorted when seen from other observer. In classical physics (Galilean relativity), the only conversion is $x = x' + vt'$ with $t = t'$.

Lorentz Transformation

In addition, we have

$$\frac{dt}{dt'} = \gamma(1 + \beta \frac{u'_x}{c}) \equiv \gamma\sigma \quad \text{with} \quad \sigma = 1 + \beta \frac{u'_x}{c}$$

Note that this gives the **time dilation**, e.g., a clock at rest in the system K' is seen in the system K with $\Delta t = \gamma \Delta t'$. Then the velocity conversion is:

$$\begin{aligned} u_x &= \frac{dx}{dt} = \frac{dx}{dt'} \frac{dt'}{dt} = \gamma(u'_x + v) \frac{dt'}{dt} = \frac{1}{\sigma} \gamma(u'_x + v) \\ u_y &= \frac{dy}{dt} = \frac{dy}{dt'} \frac{dt'}{dt} = u'_y \frac{dt'}{dt} = \frac{1}{\gamma\sigma} u'_y \\ u_z &= \frac{dz}{dt} = \frac{dz}{dt'} \frac{dt'}{dt} = u'_z \frac{dt'}{dt} = \frac{1}{\gamma\sigma} u'_z \end{aligned}$$

The transformation can be formulated in terms of the velocity components $\parallel$ and $\perp \mathbf{v}$:

$$u_{\parallel} = \frac{u'_{\parallel} + v}{1 + \beta \frac{u'_{\parallel}}{c}} \quad \text{and} \quad u_{\perp} = \frac{u'_{\perp}}{\gamma(1 + \beta \frac{u'_{\parallel}}{c})}$$

Note that $u_{\perp} \ll u'_{\perp}$ when $\gamma \gg 1$ and $\beta > 0 \implies$ **beaming effect seen in frame K**.

Lorentz Transformation

As for the acceleration, we have

$$\begin{aligned}a_{\parallel} &= \frac{du_{\parallel}}{dt} = \frac{du_{\parallel}}{dt'} \frac{dt'}{dt} = \gamma^{-3} \sigma^{-3} a'_{\parallel} \\a_{\perp} &= \frac{du_{\perp}}{dt} = \frac{du_{\perp}}{dt'} \frac{dt'}{dt} = \gamma^{-2} \sigma^{-3} \left(\sigma a'_{\perp} - \beta \frac{u'_{\perp}}{c} a'_{\parallel} \right)\end{aligned}$$

For the special case when the particle is at rest in system K' , that is, that initially $u_{\parallel} = u_{\perp} = 0$, we have $\sigma = 1$ and

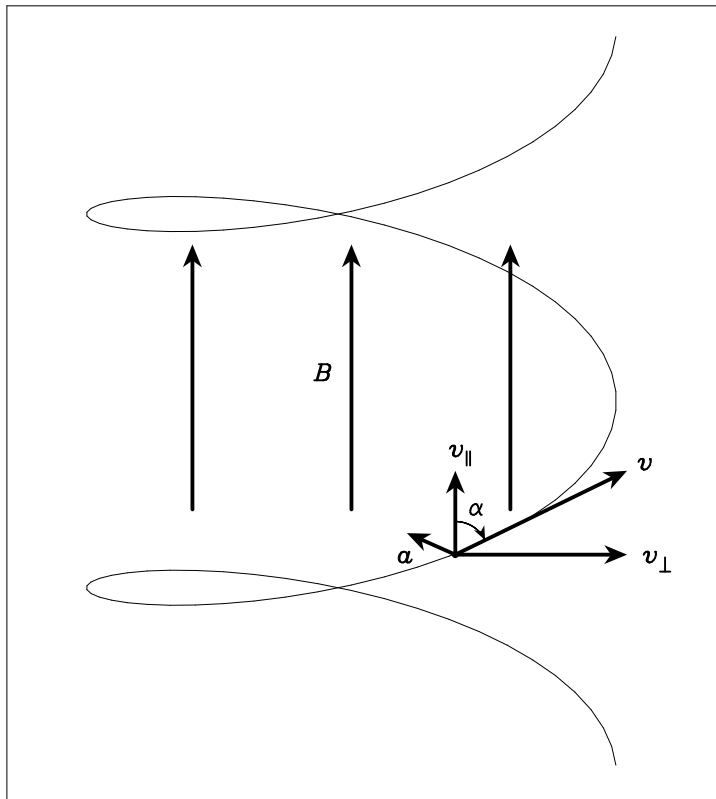
$$a_{\parallel} = \gamma^{-3} a'_{\parallel} \quad \text{and} \quad a_{\perp} = \gamma^{-2} a'_{\perp}$$

Synchrotron Radiation of 1 Electron

The motion of an electron in a (homogeneous) magnetic field is governed by ($m \Rightarrow \gamma m$)

$$\frac{d}{dt}(\gamma m \mathbf{v}) = \frac{e}{c}(\mathbf{v} \times \mathbf{B})$$

where m is the rest mass.



Projecting $\mathbf{v}$ into two components,
 $\mathbf{v}_{\parallel}$ parallel to $\mathbf{B}$ and $\mathbf{v}_{\perp}$ perpendicular to $\mathbf{B}$, we have

$$\frac{d\mathbf{v}_{\parallel}}{dt} = 0 \quad \text{and} \quad \frac{d\mathbf{v}_{\perp}}{dt} = \frac{e}{\gamma m c}(\mathbf{v}_{\perp} \times \mathbf{B})$$

So, $\mathbf{v}_{\parallel} = \text{constant}$.

Since the force is perpendicular to $\mathbf{v}_{\perp}$, the motion is a uniform circular motion with a constant orbital velocity $v_{\perp} \Leftrightarrow$ **relativistic cyclotron (gyration) radiation**. The path followed by the electron is a helix winding around $\mathbf{B}$ with the constant **pitch angle** (the angle betw. the field and velocity) $\tan \alpha = \frac{|\mathbf{v}_{\perp}|}{|\mathbf{v}_{\parallel}|}$.

Synchrotron Radiation of 1 Electron

The frequency of gyration:

$$\omega_B = \frac{eB}{\gamma mc} \equiv \frac{\omega_G}{\gamma} \quad \text{with} \quad \omega_G = \frac{eB}{mc}$$

Since the "mass" is larger by a factor of γ than the rest mass, the frequency is smaller by a factor of γ .

The acceleration is $a_{\perp} = \omega_B v_{\perp}$.

Now if we select an inertial frame K' that moves with respect to rest frame K such that the electron is at rest at a certain time, then this particle will not remain at rest for long since its acceleration is not zero. In frame K' , the electron is nonrelativistic and radiates according to the Larmor formula:

$$P' = \frac{2e^2}{3c^3} a_{\perp}'^2$$

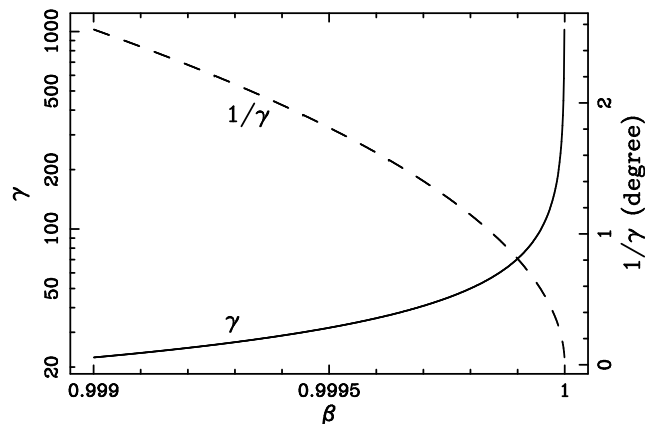
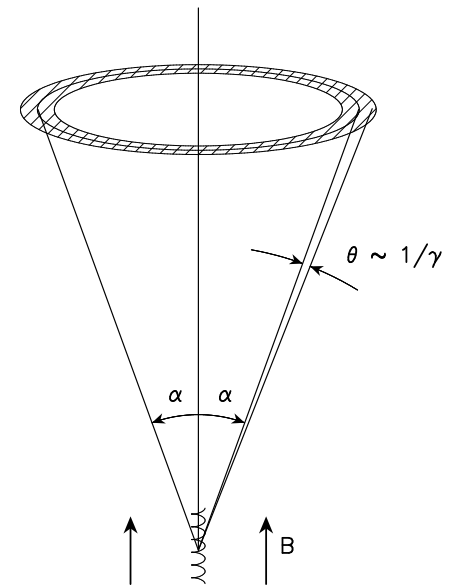
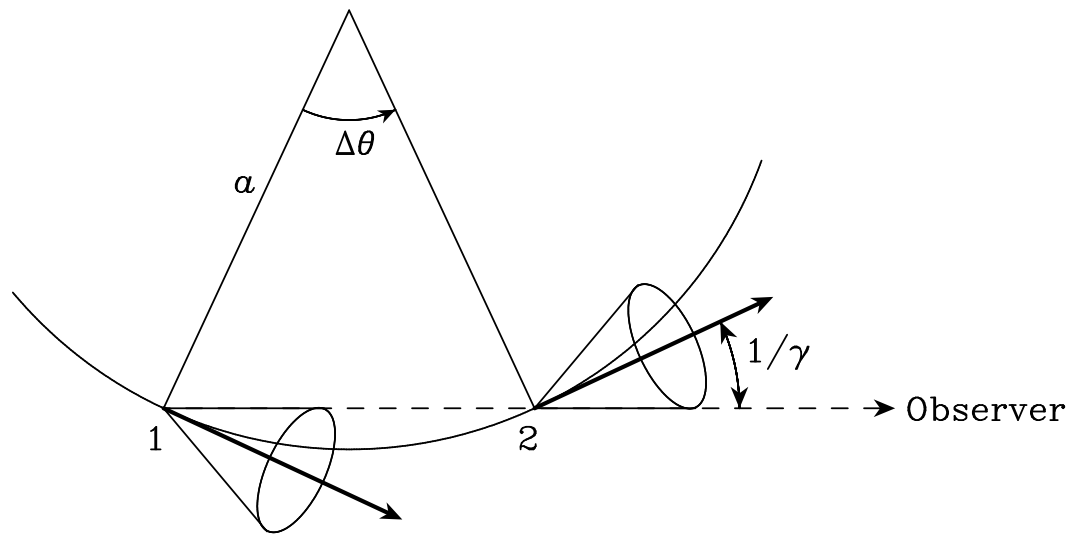
The total power radiated in frame K (observer's frame) is then (with $a'_{\perp} \Rightarrow \gamma^2 a_{\perp}$)

$$P = \frac{2e^2}{3c^3} \gamma^4 a_{\perp}^2 = \frac{2e^2}{3c^3} \gamma^4 \frac{e^2 B^2}{\gamma^2 m^2 c^2} v_{\perp}^2 = 2\sigma_T c \beta_{\perp}^2 \gamma^2 U_B$$

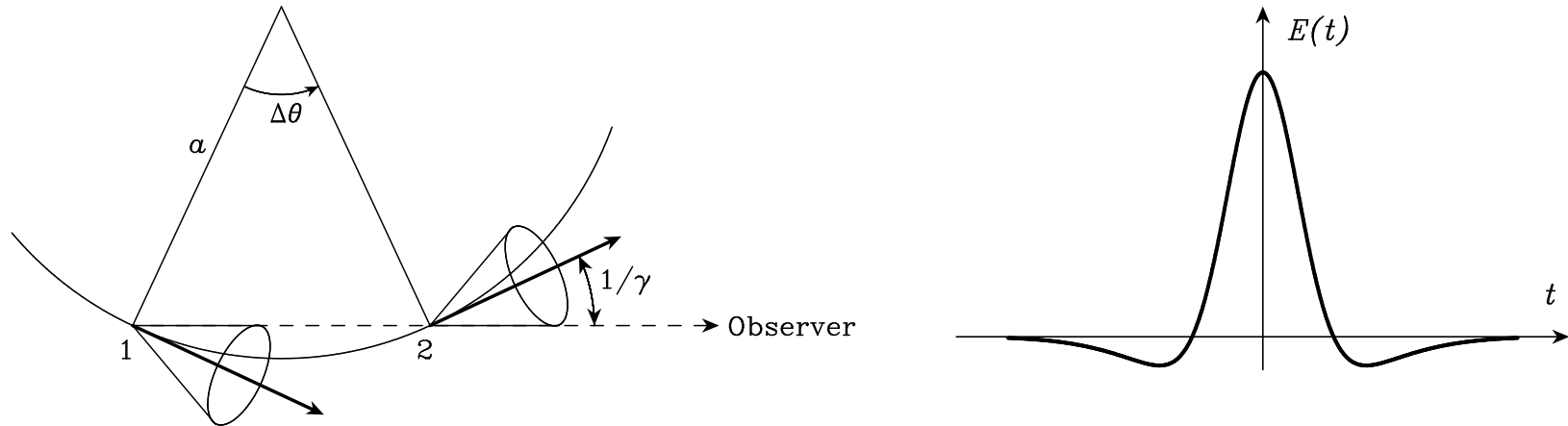
where $\sigma_T = \frac{8\pi r_0^2}{3} = 6.65 \times 10^{-25} \text{ cm}^2$ is the Thomson scattering cross section of electron with $r_0 = \frac{e^2}{mc^2}$ (classical electron radius), $\beta_{\perp} = \frac{v_{\perp}}{c}$, and magnetic energy density $U_B = \frac{B^2}{8\pi}$.

Beaming effect

In frame K' , the radiation pattern is similar to that of the Hertz dipole with the dipole (or *current* I) direction given by the acceleration direction of the electron. But in frame K (observer's frame) the radiation is shifted to the forward direction and confined to a cone with an angle of $1/\gamma$.



Beaming effect



The observer will see the pulse of radiation from points 1 and 2 along the electron's path.

$$\Delta t = t_2 - t_1 \approx \frac{\Delta\theta}{\omega_B \sin \alpha} = \frac{2/\gamma}{\omega_B \sin \alpha}$$

Let t_1^A and t_2^A be the **arrival times** of the radiation at the observer. The difference $t_2^A - t_1^A$ is less than $t_2 - t_1$ by an amount $\Delta s/c$ (**Doppler effect**) where $\Delta s = v(t_2 - t_1)$. Thus,

$$\Delta t^A = t_2^A - t_1^A = \Delta t \left(1 - \frac{v}{c}\right) = \Delta t \frac{\left(1 - \frac{v^2}{c^2}\right)}{1 + \frac{v}{c}} \approx \frac{\Delta t}{2\gamma^2} = \frac{1}{\gamma^3 \omega_B \sin \alpha}$$

Synchrotron Radiation

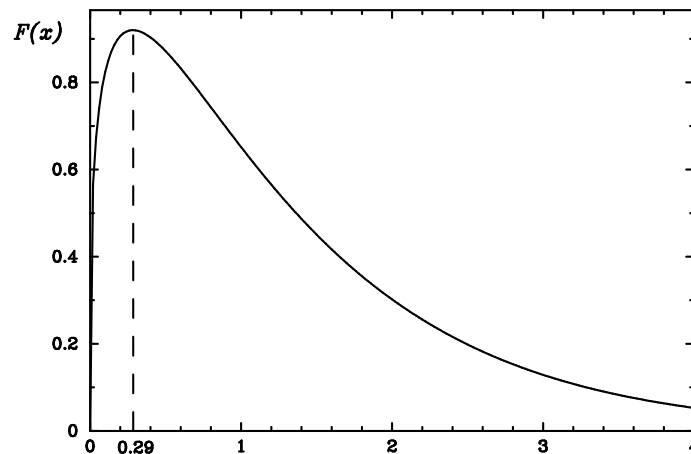
The spectrum of the radiation can be obtained by Fourier transform of the radiation pulse received. The emissivity is:

$$P(\nu) = \frac{\sqrt{3}e^3 B \sin \alpha}{mc^2} F\left(\frac{\nu}{\nu_c}\right)$$

where the critical frequency ν_c is defined by

$$\nu_c = \frac{3}{2}\gamma^3 \nu_B \sin \alpha = \frac{3}{2}\gamma^2 \nu_G \sin \alpha \quad \text{with} \quad \nu_G = \frac{\omega_G}{2\pi} = \frac{eB}{2\pi mc}$$

and $F(x) = x \int_x^\infty K_{5/3} dx$ with $K_{5/3}$ being the modified Bessel function of 5/3 order.



For an ensemble of electrons

From empirical evidence, cosmic ray data show that number density of electrons with energies between E and $E + dE$ (or γ and $\gamma + d\gamma$ since $E = \gamma mc^2$) can be approximately expressed in the form

$$N(E)dE = CE^{-p}dE \quad E_1 < E < E_2$$

or

$$N(\gamma)d\gamma = C\gamma^{-p}d\gamma \quad \gamma_1 < \gamma < \gamma_2$$

The quantity C can vary with pitch angle and the like. The emissivity (power per unit frequency interval per unit volume and per unit solid angle) of the relativistic electrons is given by

$$j_\nu = C \int_{\gamma_1}^{\gamma_2} P(\nu)\gamma^{-p}d\gamma \propto \int_{\gamma_1}^{\gamma_2} F\left(\frac{\nu}{\nu_c}\right)\gamma^{-p}d\gamma$$

Let $x \equiv \nu/\nu_c$, noting $\nu_c \propto \gamma^2$, then

$$j_\nu \propto \nu^{-(p-1)/2} \int_{x_1}^{x_2} F(x)x^{(p-3)/2}dx \propto \nu^{-(p-1)/2}$$

Putting in all the quantities we have

$$j_\nu = \frac{\sqrt{3}e^3CB \sin \alpha}{mc^2(p+1)} \Gamma\left(\frac{3p+19}{12}\right) \Gamma\left(\frac{3p-1}{12}\right) \left(\frac{2\pi mc\nu}{3eB \sin \alpha}\right)^{-(p-1)/2}$$

Absorption

Synchrotron emission is accompanied by [absorption](#), in which a photon interacts with a charge in a magnetic field and is absorbed, giving up its energy to the charge. Another process that can occur is [stimulated emission or negative absorption](#), in which a particle is induced to emit more strongly into a direction and at a frequency where photons already present. This absorption can become important in compact, high-intensity radio sources at low frequencies when the optical depth becomes large.

The derivation of the absorption coefficient is complicated. Here is the result:

$$\alpha_\nu = \frac{\sqrt{3}e^3}{8\pi m} \left(\frac{3e}{2\pi m^3 c^5} \right)^{p/2} C(B \sin \alpha)^{(p+2)/2} \Gamma\left(\frac{3p+2}{12}\right) \Gamma\left(\frac{3p+22}{12}\right) \nu^{-(p+4)/2}$$

The source function can be found from

$$S_\nu = \frac{j_\nu}{\alpha_\nu} \propto \nu^{5/2}$$

The source function is a power with an index $-\frac{5}{2}$, independent of the value of p . This index is not equal to -2 , the Rayleigh-Jeans value, because the emission is nonthermal.

Spectrum of Synchrotron Radiation

The radiative transfer can be written as

$$I_\nu(\tau_\nu) = S_\nu(1 - e^{-\tau_\nu})$$

and thus

$$I_\nu(\tau_\nu) = \begin{cases} S_\nu \propto \nu^{5/2} & \text{for } \tau_\nu \gg 1 \\ S_\nu \tau_\nu \approx j_\nu L \propto \nu^{-(p-1)/2} & \text{for } \tau_\nu \ll 1 \end{cases}$$

with L being the depth of the emission region along the line of sight.

